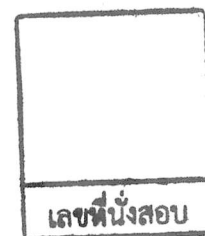


College of Industrial Technology

King Mongkut's University of Technology North Bangkok



Final Examination of Semester 2

Year 2017

Subject 394172 Mathematics 2

Section 15-18

Date 16 May 2018

Time 13.00-16.00

Name _____ ID _____ Class _____

Instructions:

1. The examination has 8 pages (including this page) and a total score of 120 points.
2. Write your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor permission.
6. You not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculations are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Question 1. solve the absolutely inequality $|x-1||2-x| \leq |x+2|$ and graph the solution in the real line. (10 points)

Question 2. Given A is a set of the solution of $\frac{3|1-x|}{x-1} < -x$ and
 B is a set of the solution of $\frac{|x|}{|x|-1} < 2$. Find $A-B$. (10 points)

Question 3. Evaluate the radical expression $\frac{1}{\sqrt{11-2\sqrt{30}}} - \frac{3}{\sqrt{7-2\sqrt{10}}} - \frac{4}{\sqrt{8-4\sqrt{3}}}$.

(10 points)

Question 4. solve the radical equation $(3+x)^{\frac{2}{3}} + 4(3-x)^{\frac{2}{3}} = 5(9-x^2)^{\frac{1}{3}}$.

(10 points)

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Question 5. 5.1 If $a \mid (-5x + 3y)$ and $a \mid (7x - 2y)$, show that $a \mid y$. (5 points)

5.2 Apply Euclidean algorithm, find the value of $\gcd(288, 616)$ and $\text{lcm}(288, 616)$. (5 points)

Question 6. find the solution set of the exponential equation

$$2 \cdot 2^{2x+1} + 6^x - 2 \cdot 3^{2x+2} = 0. \quad (10 \text{ points})$$

Question 7. Given the exponential inequality $\left(\frac{1}{\sqrt{2}}\right)^{x^2+2x+8} \leq \left(\frac{1}{2}\right)^{x+12}$. Find the solution set. (10 points)

Question 8. Given $\frac{1}{4}\log_a \frac{81}{16} - 4\log_a 2 + \frac{1}{3}\log_a 216 + \log_a \frac{3}{100} + 2\log_a 5 + 6\log_a 2 = 6$.

Evaluate the value of a^2 . (10 points)

Question 9. Find all solutions of the logarithmic inequality

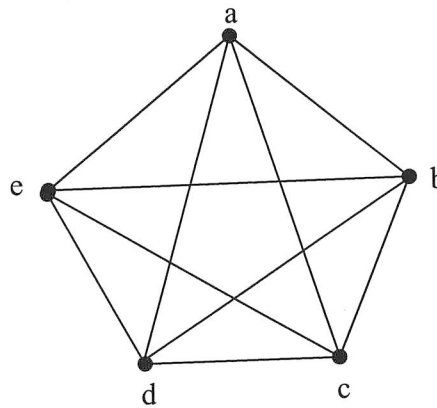
$$\log_{0.5}(3x-4) \geq \log_{0.5}(x^2-2).$$

(10 points)

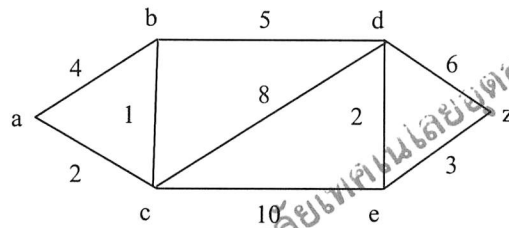
Question 10. Find the product of x and y of two expo-logarithm equations

$$\begin{aligned} 9^{\log_3 x} + 4^{\log_{10} y} &= 16, \\ \log_6 y \log_8 6 \log_{10} 8 &= 2. \end{aligned} \quad (10 \text{ points})$$

Question 11. Find the number of simple paths between the vertex a and the vertex d in the graph of length 3 and show that this connected multigraph has Euler circuit or Euler path whether or not. (10 points)



Question 12. What is the shortest path between the vertex **a** and the vertex **z** in the weighted graph? (10 points)



Assoc. Prof. Dr. Sanoe Koonprasert
and Dr. Sekson Sirisubtawee.